



NCIE-013

Functional Modules Specification

**In development — for human review of full expansion —
Version 1.0**

Specifies domain business-rule, workflow and analytical-product logic beneath the approved NCIE-001–012 foundation: cross-domain functional doctrine, module lifecycle, rule/KPI/anomaly/alert frameworks, case handoff, cross-domain correlation, the eight domain intelligence modules (Network, Traffic, Revenue, Mobile Money, SIM, Anti-Fraud, REWS, Operator/Entity Profile), geospatial and temporal/forecast interpretation, domain Evidence integration, ARGUS/Agent use, workflow, API/UX consumption, security, explainability, data quality, configuration governance, degraded operation and domain testing.

Domain-only, upstream-consuming: NCIE-013 owns domain business rules, workflows and analytical products; it does not redefine canonical data, ARGUS behavior, Memory/Context, Agent/VPF semantics, governance, IAM/security, Evidence/provenance, API contracts or UX semantics.

Expands approved NCIE-013 Skeletal v0.1, itself traced to approved NCIE-001 v1.3 through approved NCIE-012 v1.0.

Prepared: 15 September 2026

Maintained by: National Communications Intelligence Ecosystem (NCIE)

Document Governance

0.1 Formal-Title Reconciliation (Directive Item 3)

The approved NCIE-013 Skeletal v0.1 carries the working title 'NCIE-013 — Domain Functional Modules Specification' and directs that the formal title be reconciled against the Master Controlled Register if its wording differs. This pre-approval surgical correction locates and applies that reconciliation: the authoritative source is the NCIE Documentation Suite Master Controlled Register v2.0 (FOR APPROVAL, dated 02 September 2026), Section 3 'Reconciled Master Documentation Register', which records the NCIE-013 entry as 'Functional Modules Specification' with Primary Purpose 'Detailed specifications for Network, Traffic, Revenue, Mobile Money, SIM Registration, Anti-Fraud, REWS and related operational modules' — matching this document's actual Chapters 11–18 domain-module scope exactly. Comparison result: SEMANTIC MATCH — DOCUMENT-CONTROL WORDING DIFFERS (the register's title omits 'Domain'; no material discrepancy in scope or substance was found). Per this result, the document-control title wording has been corrected throughout this document (cover, this section, Document Governance table, Authority Chain, running header, Chapter 34 disposition statement and core document properties) to 'NCIE-013 — Functional Modules Specification'; no chapter content, module architecture, rule/KPI framework, or other approved substance is altered by this correction. This correction, and the prior source-availability defect it corrects, are recorded in the Full Expansion Change Register and remain cross-referenced as Human Review item HR13-1-1, which stays open pending institutional confirmation of the NCIE-013 acceptance/sign-off authority and module ownership.

0.2 Authority Chain

NCIE-001 requirements → NCIE-002 architecture → NCIE-003 canonical data → NCIE-004 engineering baseline → NCIE-005 ARGUS behavioral authority → NCIE-006 Memory & Context architecture → NCIE-007 AI/Agent/Model Orchestration → NCIE-008 Governance, Authority, Accountability & RACI → NCIE-009 Security, IAM & Zero-Trust Architecture → NCIE-010 Evidence, Provenance & Audit Architecture → NCIE-011 API, Integration & Data Connector Specification → NCIE-012 UX/UI & Design System Specification → NCIE-013 Functional Modules Specification. NCIE-013 interprets governed domain data through upstream semantics and authority; it creates no canonical meaning, institutional authority, security control, Evidence semantics, API contract or UX pattern of its own.

0.3 Token-Conservation Convention

Every chapter opens with a short Upstream block citing the specific upstream NCIE-001–012 authority the chapter interprets, stating only the domain-logic relationship rather than restating upstream technical, governance, security, Evidence, integration or UX content.

0.4 Governing Functional-Module Invariants Register

These 18 invariants bind every chapter of this document and are stated once here rather than re-derived repeatedly. They are fixed by the approved NCIE-013 Skeletal v0.1 and by the approved NCIE-001–012 authority chain; no institutional decision is required to adopt them. The 12 NCIE-008 governance invariants, the 16 NCIE-009 security invariants, the 19 NCIE-010 Evidence/provenance/audit invariants, the 20 NCIE-011 integration invariants, the 19 NCIE-012 UX/UI invariants, and other upstream invariants already binding from NCIE-005/006/007 remain fully in force and are cross-referenced, not restated, in the chapters below.

- DOMAIN MODULE ≠ SYSTEM OF RECORD.
- DOMAIN LOGIC ≠ CANONICAL DATA OWNERSHIP.
- ANALYTICAL SIGNAL ≠ EVIDENCE.
- ALERT ≠ FINDING.
- SITUATION ≠ INSTITUTIONAL DECISION.
- KPI THRESHOLD BREACH ≠ PROVEN VIOLATION.
- CORRELATION ≠ CAUSATION.
- ANOMALY ≠ FRAUD.
- RISK SCORE ≠ GUILT / LIABILITY.
- MODEL OUTPUT ≠ REGULATORY FINDING.
- VPF PASS ≠ DOMAIN APPROVAL.

- ARGUS ASSISTANCE ≠ DOMAIN AUTHORITY.
- WORKFLOW STATE ≠ HUMAN APPROVAL.
- CONNECTOR AVAILABILITY ≠ SOURCE TRUSTWORTHINESS.
- MISSING DATA ≠ ZERO VALUE.
- LATE DATA ≠ CURRENT TRUTH.
- CROSS-DOMAIN CORRELATION ≠ CROSS-DOMAIN DISCLOSURE AUTHORITY.
- AUTOMATION ≠ UNBOUNDED CONSEQUENTIAL ACTION.

0.5 Domain/Upstream Boundary Note

NCIE-013 owns domain business rules, workflows and analytical products. It consumes NCIE-003 canonical data, NCIE-005/006/007 ARGUS/Memory/orchestration semantics, NCIE-008 institutional decision rights, NCIE-009 security/IAM controls, NCIE-010 Evidence/provenance/audit semantics, NCIE-011 API/integration contracts and NCIE-012 UX patterns, without redefining any of them. Where a matter requires a regulatory, policy or domain-data choice (a KPI catalogue, a fraud typology, a materiality threshold) not fixed by an approved source, this document states the analytical framework and defers the choice to Human Review rather than inventing an answer. Approval of the NCIE-013 Skeletal v0.1 authorized this full expansion only; it did not itself approve any rule catalogue, threshold, typology or ownership assignment proposed below.

0.6 Human Review Focus Classification

Every chapter's Human Review Focus item carries one of seven labels: RESOLVED BY UPSTREAM BASELINE; PROPOSED DESIGN DEFAULT; INSTITUTIONAL CONFIRMATION REQUIRED; SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED; LEGAL/POLICY CONFIRMATION REQUIRED; ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED; or SOURCE/WORKFLOW DISCOVERY REQUIRED — each additionally marked Blocking or Non-Blocking, and each stating precisely what it blocks. Per directive item 60, a Chapter 33 register item that restates an identical NCIE-008–012 governance, security, integration or UX question cross-references that upstream Human Review ID instead of duplicating it; a new NCIE-013 item is created only for a distinct domain-business decision. Chapter 33 consolidates every open item, extracted directly from the chapters so the register cannot drift from its sources (mechanically verified per §33.4).

0.7 Status Convention

Skeletal-structure approval authorizes expansion; it does not approve any individual rule catalogue, threshold, typology or ownership assignment proposed below — those remain IN DEVELOPMENT / FOR HUMAN REVIEW until a separate approval act. Nothing in this document may be cited as an approved institutional decision, a calibrated model, a validated rule, or a completed test.

0.8 Salient Things to Note Discipline

Directive item 71 requires this expansion to programmatically verify an explicit '<N>.0 Salient Things to Note' subsection in all 34 chapters, continuing the discipline established after the NCIE-009 Chapter-27 omission (found by independent human examination and corrected) and reinforced in every full expansion since. Every chapter below renders its Salient Things to Note subsection as an unconditional part of the same build-time helper (NCIE-013's lib.py raises a build error if a chapter's salient content is missing), and Chapter 34's mechanical audit (§34.2) re-verifies this against the final document.

0.9 Document Amendment History

Version: v0.1

Change: Approved skeletal review edition (structural/conceptual review only; full expansion not yet authorized).

Version: v1.0

Change: Full expansion of the approved NCIE-013 Skeletal v0.1 into the complete 34-chapter specification per the NCIE-013 v0.1→v1.0 Claude Full-Expansion Production Directive: cross-domain functional doctrine, five consolidated matrices, threshold/rule-fabrication and Finding/Decision audits, Human Review Register (Chapter 33) and full traceability/handover (Chapter 34). Formal title corrected per §0.1 pre-approval surgical correction: NCIE Documentation Suite Master Controlled Register v2.0 located and applied; title reconciled to 'NCIE-013 — Functional Modules Specification' (document-control wording only).

Authority: Approved NCIE-001 Master PRD v1.3 through approved NCIE-012 UX/UI & Design System Specification v1.0, and the approved NCIE-013 Skeletal v0.1.

Scope: Controlled full expansion only — the approved 34-chapter skeletal structure, chapter purposes and boundaries are preserved exactly; no chapter was added, removed, merged, split or renumbered.

Status: v1.0 IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION. The approved v0.1 skeleton remains historically preserved as the expansion architecture.

Document	NCIE-013 Functional Modules Specification
Edition	In development — Version 1.0 (Full Expansion of Approved Skeletal v0.1)
Requirements authority	NCIE-001 Master Product Requirements Document, Version 1.3 — APPROVED
Architecture authority	NCIE-002 System Architecture & Technical Design, Version 0.5 — APPROVED
Canonical data authority	NCIE-003 Data Model & Data Dictionary, Version 0.4 — APPROVED
Engineering baseline	NCIE-004 Technology Stack & Engineering Blueprint, Version 1.2 — APPROVED
ARGUS behavioral authority	NCIE-005 ARGUS Intelligence Assistant Specification, Version 1.3 — APPROVED
Memory & Context authority	NCIE-006 Memory & Context Architecture Specification, Version 1.2 — APPROVED
Orchestration authority	NCIE-007 AI, Agent & Model Orchestration Specification, Version 1.2 — APPROVED
Governance authority	NCIE-008 Governance, Authority, Accountability & RACI Specification, Version 1.0 — APPROVED
Security authority	NCIE-009 Security, IAM & Zero-Trust Architecture, Version 1.0 — APPROVED
Evidence/provenance/audit authority	NCIE-010 Evidence, Provenance & Audit Architecture, Version 1.0 — APPROVED
Integration authority	NCIE-011 API, Integration & Data Connector Specification, Version 1.0 — APPROVED
UX/UI authority	NCIE-012 UX/UI & Design System Specification, Version 1.0 — APPROVED
Prior baseline	NCIE-013 Functional Modules Specification, Skeletal v0.1 — APPROVED FOR FULL EXPANSION (historically preserved)
Status	IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION until explicitly approved; the approved v0.1 skeleton remains historically preserved as the expansion architecture

Contents

This table of contents updates automatically in Microsoft Word. If page numbers do not appear immediately, select the field and press F9, or use References → Update Table.

0.1 Formal-Title Reconciliation (Directive Item 3)	2
0.2 Authority Chain	2

0.3 Token-Conservation Convention	2
0.4 Governing Functional-Module Invariants Register	2
0.5 Domain/Upstream Boundary Note.....	3
0.6 Human Review Focus Classification	3
0.7 Status Convention	3
0.8 Salient Things to Note Discipline	3
Mandate, Authority Chain & Domain-Module Boundary	11
1.0 Salient Things to Note	11
1.1 Mandate and Scope.....	11
1.2 Domain-Module Boundary — What NCIE-013 Does Not Own	11
1.3 Handovers	11
1.R RACI / Control Ownership	11
Human Review Focus	12
Cross-Domain Functional Doctrine	13
2.0 Salient Things to Note	13
2.1 Analytical States Remain Analytical.....	13
2.2 Uncertainty, Quality and Explainability	13
2.3 Domain-State Matrix (directive item 54)	13
2.R RACI / Control Ownership	14
Human Review Focus	14
Module Architecture & Common Lifecycle	15
3.0 Salient Things to Note	15
3.1 Common Lifecycle	15
3.2 Common Lifecycle Exceptions	15
3.3 Module Catalogue Matrix (directive item 55)	15
3.R RACI / Control Ownership	17
Human Review Focus	17
Domain Data Inputs & Canonical Mapping.....	18
4.0 Salient Things to Note	18
4.1 Domain Logic Does Not Own Canonical Data	18
4.2 Quality, Temporal Fields and Missing/Late Data	18
4.R RACI / Control Ownership	18
Human Review Focus	18
Domain Rule & Policy Evaluation Framework	19
5.0 Salient Things to Note	19
5.1 Triggered Rule Is Not Violation.....	19
5.2 Rule Identity and Provenance	19
5.R RACI / Control Ownership	19
Human Review Focus	19
KPI, Threshold & Baseline Framework.....	20
6.0 Salient Things to Note	20

6.1 Breach Is Not Proven Violation	20
6.2 KPI Catalogue and Thresholds.....	20
6.3 Rule/KPI Governance Matrix (directive item 56)	20
6.R RACI / Control Ownership	21
Human Review Focus	21
Anomaly, Risk & Scoring Framework.....	22
7.0 Salient Things to Note	22
7.1 Anomaly Is Not Fraud, Score Is Not Guilt.....	22
7.2 Prohibited Autonomous Consequences	22
7.R RACI / Control Ownership	22
Human Review Focus	22
Alert, Situation & Escalation Framework	23
8.0 Salient Things to Note	23
8.1 Alert Is Not Finding	23
8.2 Acknowledgement, Escalation, Suppression and Closure.....	23
8.R RACI / Control Ownership	23
Human Review Focus	23
Case / Investigation Handoff.....	24
9.0 Salient Things to Note	24
9.1 Analytics Cannot Autonomously Create Institutional Findings	24
9.2 Referral Provenance	24
9.R RACI / Control Ownership	24
Human Review Focus	24
Cross-Domain Correlation & Fusion	25
10.0 Salient Things to Note	25
10.1 Correlation Is Not Causation or Disclosure Authority	25
10.2 Contradictions and Purpose.....	25
10.3 Cross-Domain Correlation Matrix (directive item 57)	25
10.R RACI / Control Ownership	25
Human Review Focus	26
Network Performance & Quality Module	27
11.0 Salient Things to Note	27
11.1 Degradation Is Not Breach	27
11.2 Trend and Geography.....	27
11.R RACI / Control Ownership	27
Human Review Focus	27
Traffic & Usage Intelligence Module.....	28
12.0 Salient Things to Note	28
12.1 Anomaly Is Not Misconduct	28
12.2 Classification Specifics	28
12.R RACI / Control Ownership	28

Human Review Focus	28
Revenue & Financial Assurance Module	29
13.0 Salient Things to Note	29
13.1 Variance Is Not Fraud or Liability	29
13.2 Source Precedence and Materiality	29
13.R RACI / Control Ownership	29
Human Review Focus	29
Mobile Money Intelligence Module	30
14.0 Salient Things to Note	30
14.1 Anomaly Is Not Fraud, Privacy Is Strict	30
14.2 Permitted Purposes	30
14.R RACI / Control Ownership	30
Human Review Focus	30
SIM / Subscriber Intelligence Module.....	31
15.0 Salient Things to Note	31
15.1 Linkage Is Not Wrongdoing.....	31
15.2 Identity-Use Purposes	31
15.R RACI / Control Ownership	31
Human Review Focus	31
Anti-Fraud & Abuse Intelligence Module	32
16.0 Salient Things to Note	32
16.1 Indicators Support Review, Not Accusation	32
16.2 Typology and Score Governance.....	32
16.R RACI / Control Ownership	32
Human Review Focus	32
REWS / Early-Warning Module.....	33
17.0 Salient Things to Note	33
17.1 Early Warning Is Anticipatory, Not a Finding	33
17.2 Indicator Catalogue and Escalation	33
17.R RACI / Control Ownership	33
Human Review Focus	33
Operator / Entity Profile & Comparative Intelligence	34
18.0 Salient Things to Note	34
18.1 Rank Is Not Wrongdoing.....	34
18.2 Peer-Group and Benchmark Governance	34
18.R RACI / Control Ownership	34
Human Review Focus	34
Geospatial & Topology Intelligence.....	35
19.0 Salient Things to Note	35
19.1 Spatial Relation Is Not Causality.....	35
19.2 Sensitive Layers	35

19.R RACI / Control Ownership 35

Human Review Focus 35

Temporal, Trend & Forecast Interpretation 36

20.0 Salient Things to Note 36

20.1 Forecast Is Not Fact..... 36

20.2 Late/Revised Data and Decision-Time Comparison 36

20.R RACI / Control Ownership 36

Human Review Focus 36

Domain Evidence & Provenance Integration 37

21.0 Salient Things to Note 37

21.1 Analytical Signal Remains Non-Evidence 37

21.2 Provenance, Contradictions and Decision-Time Packages..... 37

21.3 Domain Evidence Matrix (directive item 58) 37

21.R RACI / Control Ownership 38

Human Review Focus 38

ARGUS Domain Assistance Boundary..... 39

22.0 Salient Things to Note 39

22.1 Assistance Is Not Authority 39

22.2 Provenance and Limits 39

22.R RACI / Control Ownership 39

Human Review Focus 39

Agent & AI Use Within Domain Modules 40

23.0 Salient Things to Note 40

23.1 Machine Output Never Becomes Finding or Decision..... 40

23.2 Approved Domain AI/Agent Uses 40

23.R RACI / Control Ownership 40

Human Review Focus 40

Domain Workflow & Human Review Integration 41

24.0 Salient Things to Note 41

24.1 Workflow State Is Not Approval 41

24.2 Review Queue and Dissent..... 41

24.R RACI / Control Ownership 41

Human Review Focus 41

Domain API & Connector Integration 42

25.0 Salient Things to Note 42

25.1 Availability Is Not Truth 42

25.2 Standard versus Bespoke Connectors 42

25.R RACI / Control Ownership 42

Human Review Focus 42

Domain UX & Workspace Integration 43

26.0 Salient Things to Note 43

26.1 Rendering Without Redefining	43
26.2 Module Workspace Layouts	43
26.R RACI / Control Ownership	43
Human Review Focus	43
Security, Privacy & Purpose Limitation	44
27.0 Salient Things to Note	44
27.1 Analytical Value Does Not Override Purpose	44
27.2 High-Risk Access Classes	44
27.R RACI / Control Ownership	44
Human Review Focus	44
Domain Audit, Explainability & Defensibility	45
28.0 Salient Things to Note	45
28.1 Explainability Supports Review, Not Evidence	45
28.2 Minimum Explainability	45
28.R RACI / Control Ownership	45
Human Review Focus	45
Data Quality, Missingness & Reconciliation	46
29.0 Salient Things to Note	46
29.1 Missing Is Not Zero, Late Is Not Current	46
29.2 Conflict, Revision and Qualification	46
29.R RACI / Control Ownership	46
Human Review Focus	46
Domain Configuration & Change Governance	47
30.0 Salient Things to Note	47
30.1 Configuration Changes Alter Outcomes	47
30.2 Impact Assessment	47
30.R RACI / Control Ownership	47
Human Review Focus	47
Domain Performance, Scale & Degraded Operation	48
31.0 Salient Things to Note	48
31.1 Degradation Does Not Relax Governed Rules	48
31.2 Recovery and Reconciliation	48
31.R RACI / Control Ownership	48
Human Review Focus	48
Functional Module Testing & Scenario Validation	49
32.0 Salient Things to Note	49
32.1 Test Classes	49
32.2 Test-Status Integrity	49
32.R RACI / Control Ownership	50
Human Review Focus	50
Human Review Register & Open Domain Decisions	51

33.0 Salient Things to Note 51

33.1 Purpose 51

33.2 Summary 51

33.3 Complete Register..... 51

33.4 Mechanical Verification 56

33.5 Upstream Deduplication (Directive Item 60)..... 56

33.6 Blocking Scope Discipline 56

Human Review Focus 57

Traceability, Acceptance & Downstream Handover 58

34.0 Salient Things to Note 58

34.1 Traceability Matrix 58

34.2 Chapter-Count and Salient-Note Mechanical Verification 59

34.3 Skeleton-to-Full Fidelity Verification 59

34.4 Acceptance Criteria and the Unresolved-Item Gate 59

34.5 Downstream Handover Register 60

34.6 Readiness and Final Disposition 60

Human Review Focus 60

Mandate, Authority Chain & Domain-Module Boundary

Establish NCIE-013 ownership of domain functional-module business logic within the approved NCIE suite.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-001–012; NCIE-005/011/012 handovers.

1.0 Salient Things to Note

Interpret governed domain data without redefining upstream semantics or authority — Table 1 fixes this boundary so no later chapter can drift into owning what an upstream document already owns.

1.1 Mandate and Scope

NCIE-013 interprets governed domain data through upstream semantics and authority — it does not create product behavior, canonical meaning, institutional authority, security policy, Evidence semantics, API contracts or UX patterns.

1.2 Domain-Module Boundary — What NCIE-013 Does Not Own

NCIE-013 does not own canonical data, ARGUS behavior, Memory/Context semantics, Agent/VPF semantics, institutional governance, security enforcement, Evidence/provenance meaning, API contracts or UX semantics. It owns the domain business-rule, workflow and analytical-product layer above all of them.

Table 1. NCIE-013 domain-module boundary.

Concern	Owned By
Product requirements/scope	NCIE-001
System architecture	NCIE-002
Canonical data model	NCIE-003
Engineering baseline	NCIE-004
ARGUS behavioral authority	NCIE-005
Memory & Context architecture	NCIE-006
AI/Agent/Model orchestration	NCIE-007
Institutional governance, RACI	NCIE-008
Security, IAM, Zero-Trust enforcement	NCIE-009
Evidence lifecycle, provenance, audit	NCIE-010
API/event/connector/integration contracts	NCIE-011
UX/UI, presentation, interaction, Design System	NCIE-012
Domain business rules, workflows, analytical products	NCIE-013 (this document)
Deployment/operations, detailed testing	NCIE-015/016 (downstream)

1.3 Handovers

NCIE-005 excludes functional-module detail to NCIE-013; NCIE-011 hands over domain connector patterns (Ch.25); NCIE-012 hands over domain-workflow UX patterns (Ch.26). Every consequential domain state in this document renders an upstream state — never a new institutional authority invented here.

1.R RACI / Control Ownership

Table 2. Chapter 1 RACI / control ownership.

Function	Accountable (NCIE-	Responsible	Consulted	Informed
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	008)	(Domain)		
Adopt/confirm	Accountable	Domain	NCIE-	All institutional role
NCIE-013 domain	Authority Class	Architecture Owner	008/009/010/011/012	holders
architecture	(NCIE-008, to be assigned)	(to be assigned)	chapter owners	

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-1-1: Confirm the NCIE-013 acceptance/sign-off authority and module ownership. The formal title has been reconciled against the NCIE Documentation Suite Master Controlled Register v2.0 (FOR APPROVAL) §3 — entry 'Functional Modules Specification' — and the document-control title corrected accordingly (§0.1); this item remains open pending institutional confirmation of acceptance authority and module ownership, and pending separate approval of the Master Controlled Register v2.0 itself.

Traceability: NCIE-001–012; NCIE-005/011/012 handovers.

Cross-Domain Functional Doctrine

Define common business-rule principles governing every domain module this document specifies.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-001/003/008/010.

2.0 Salient Things to Note

Signals, alerts, anomalies and scores remain analytical states until governed processes establish Findings/Decisions — Table 3 fixes the complete state hierarchy and an authority caution for each state so no chapter can present one state as if it were a later one.

2.1 Analytical States Remain Analytical

Signals, alerts, anomalies and scores remain analytical states until a governed institutional process establishes a Finding or Decision (NCIE-008 Ch.18) — no domain module's output becomes a Finding merely by being generated, however confident or precisely calculated.

2.2 Uncertainty, Quality and Explainability

Every domain output carries its data-quality context, its uncertainty where applicable, and a path to explanation (Chapter 28) — a clean-looking number is never presented without the quality/completeness context that would let a reviewer judge how much to trust it.

2.3 Domain-State Matrix (directive item 54)

The consolidated matrix below distinguishes every domain state from observation through Decision, with an authority/provenance caution for each, so no later chapter needs to re-derive this hierarchy.

Table 3. Domain-State Matrix.

State	Meaning	Authority/Provenance Caution	Owning Chapter
Observation	Raw or lightly processed domain data as received	DOMAIN LOGIC ≠ CANONICAL DATA OWNERSHIP — the observation is not itself the canonical record	Ch.4
Analytical signal	A derived value, pattern or indicator from domain logic	ANALYTICAL SIGNAL ≠ EVIDENCE — a signal is not registered Evidence until NCIE-010 qualification	Ch.4-7
Anomaly / risk score	A statistical or model-derived deviation or score	ANOMALY ≠ FRAUD; RISK SCORE ≠ GUILT/LIABILITY	Ch.7
Alert	A triggered notification from a rule, threshold or score	ALERT ≠ FINDING	Ch.8
Situation	A composite, elevated awareness state (often from REWS/correlation)	SITUATION ≠ INSTITUTIONAL DECISION	Ch.8, 17
Evidence (domain-linked)	A registered NCIE-010 Evidence Object referenced by domain analysis	Evidence status comes only through the NCIE-010 lifecycle, never by domain signal alone	Ch.21

Case / Investigation candidate	A referral into a governed case/investigation workflow	A candidate is not itself an opened case until NCIE-008/NCIE-013 Ch.9 governed acceptance	Ch.9
Finding	A formal institutional finding issued under NCIE-008 authority	Never created autonomously by a domain module — always an NCIE-008 Ch.18 act	NCIE-008 (external)
Decision	A formal institutional decision issued under NCIE-008 authority	Never created autonomously by a domain module — always an NCIE-008 Ch.18 act	NCIE-008 (external)

2.R RACI / Control Ownership

Table 4. Chapter 2 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve common domain-state vocabulary	Accountable Authority Class (to be assigned)	Cross-Domain Doctrine Owner (to be assigned)	NCIE-008/010 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-2-1: Approve common domain-state vocabulary.

Traceability: NCIE-001/003/008/010.

Module Architecture & Common Lifecycle

Define the reusable module lifecycle shared across domain modules.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002/011/012.

3.0 Salient Things to Note

Common lifecycle must not erase domain-specific semantics or approvals — Table 5 maps every module to its own rule/Evidence/security basis so the common lifecycle is a shared skeleton, never a substitute for a module's own requirements.

3.1 Common Lifecycle

Every domain module follows the same nine-stage lifecycle: observe → validate → contextualize → analyze → correlate → alert → investigate → review → close/reopen. The common lifecycle must not erase a domain-specific semantic or approval step a particular module requires — modules may extend, never bypass, the common stages.

3.2 Common Lifecycle Exceptions

Which modules require an additional lifecycle stage or a stricter gate between stages is a Human Review decision, informed by the module-specific chapters (11-18).

3.3 Module Catalogue Matrix (directive item 55)

The consolidated catalogue below maps each domain module to its inputs, outputs, rule/KPI basis, Alert/Situation classes, Evidence interface, ARGUS/Agent use, API/connector basis, UX surface, security basis and Human Review status.

Table 5. Module Catalogue Matrix.

Module	Inputs	Rules/ KPIs	Alerts/Situati ons	Evide nce Interf ace	ARGUS/ Agent Use	APIs/Conn ectors	UX	Securi ty Basis	HR Stat us
Network Performa nce & Quality (Ch.11)	Coverage/availabi lity/quality feeds	Ch.5-6 rule/KP I frame work	Degradation/o utage alerts (Ch.8)	Ch.21 doma in Evide nce	Ch.22-23	Ch.25	Ch. 26	Ch.27 (NCIE- 009)	Hol der — Hu man Revi ew
Traffic & Usage Intelligen ce (Ch.12)	Volume/utilization feeds	Ch.5-6	Volume- anomaly alerts (Ch.8)	Ch.21	Ch.22-23	Ch.25	Ch. 26	Ch.27	Hol der — Hu man Revi ew
Revenue & Financial Assuranc e (Ch.13)	Revenue/reconcili ation feeds	Ch.5-6	Leakage- indicator alerts (Ch.8)	Ch.21	Ch.22-23	Ch.25	Ch. 26	Ch.27	Hol der — Hu man Revi ew

Mobile Money Intelligence (Ch.14)	Transaction feeds (high sensitivity)	Ch.5-6	Velocity/concentration alerts (Ch.8)	Ch.21	Ch.22-23	Ch.25	Ch.26	Ch.27 (strict purpose limitation)	Holder — Human Review
SIM/Subscriber Intelligence (Ch.15)	SIM/subscriber lifecycle feeds	Ch.5-6	Pattern/exception alerts (Ch.8)	Ch.21	Ch.22-23	Ch.25	Ch.26	Ch.27 (protected identity)	Holder — Human Review
Anti-Fraud & Abuse Intelligence (Ch.16)	Cross-feed indicators	Ch.5-6 + typology reference	Score/referral alerts (Ch.8)	Ch.21	Ch.22-23	Ch.25	Ch.26	Ch.27	Holder — Human Review
REWS/Early-Warning (Ch.17)	Leading indicators across modules	Ch.5-6	Composite Situations (Ch.8)	Ch.21	Ch.22-23	Ch.25	Ch.26	Ch.27	Holder — Human Review
Operator/Entity Profile & Comparative Intelligence (Ch.18)	Normalized cross-module KPIs	Ch.5-6	Outlier/benchmark notices (Ch.8)	Ch.21	Ch.22-23	Ch.25	Ch.26	Ch.27	Holder — Human Review
Geospatial & Topology Intelligence (Ch.19)	Geo/topology layers	Ch.5-6 (where applicable)	N/A (supporting layer)	Ch.21	Ch.22-23	Ch.25	Ch.26	Ch.27 (sensitive layers)	Holder — Human Review
Temporal, Trend & Forecast Interpretation (Ch.20)	Time-series across modules	Ch.5-6 (where applicable)	Trend-break notices (Ch.8, 17)	Ch.21	Ch.22-23	Ch.25	Ch.26	Ch.27	Holder — Human Review

3.R RACI / Control Ownership

Table 6. Chapter 3 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve common-lifecycle exceptions	Accountable Authority Class (to be assigned)	Module Architecture Owner (to be assigned)	Module-specific owners (Ch.11-20)	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Non-Blocking) HR13-3-1: Confirm common lifecycle exceptions; does not block the nine-stage baseline lifecycle already defined in the interim.

Traceability: NCIE-002/011/012.

Domain Data Inputs & Canonical Mapping

Define module consumption of canonical data and connectors.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003/010/011.

4.0 Salient Things to Note

DOMAIN LOGIC ≠ CANONICAL DATA OWNERSHIP; missing/late data remain explicit — this chapter's mapping layer never introduces a silently-defaulted value where the source data was actually missing or late.

4.1 Domain Logic Does Not Own Canonical Data

DOMAIN LOGIC ≠ CANONICAL DATA OWNERSHIP. Every domain module maps its inputs to NCIE-003 canonical datasets and NCIE-011 connector contracts; a module's own transformation or derivation never becomes a competing canonical definition.

4.2 Quality, Temporal Fields and Missing/Late Data

Every mapped input carries its quality indicators, temporal fields (observed/effective/registration time) and provenance reference; missing and late data remain explicitly flagged rather than silently defaulted (Chapter 29 expands this). Source precedence, where multiple sources could supply the same field, is a Human Review decision where not already fixed upstream.

4.R RACI / Control Ownership

Table 7. Chapter 4 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve datasets/source precedence	Accountable Authority Class (to be assigned)	Domain Data Mapping Owner (to be assigned)	NCIE-003 canonical data owner, NCIE-011 connector owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-4-1: Confirm datasets and source precedence per module.

Traceability: NCIE-003/010/011.

Domain Rule & Policy Evaluation Framework

Define traceable domain rule evaluation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-003/008.

5.0 Salient Things to Note

Triggered rule ≠ violation/Finding; rules are versioned and governed — this chapter's rule-evaluation output is always labeled as an analytical outcome, never as an established violation.

5.1 Triggered Rule Is Not Violation

A rule triggering does not itself establish a violation or a Finding — every rule is versioned, carries an effective-date range, and its evaluation produces an outcome that remains an analytical state (Chapter 2) until a governed institutional process reviews it.

5.2 Rule Identity and Provenance

Every rule carries an identity, version, effective dates, its inputs, its outcomes, and any exceptions applied, with full evaluation provenance. Rule-authoring and approval authority is a Human Review decision.

5.R RACI / Control Ownership

Table 8. Chapter 5 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve rule-authoring/approval authority	Accountable Authority Class (to be assigned)	Rule Framework Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-5-1: Confirm rule-authoring/approval authority.
- Traceability: NCIE-003/008.

KPI, Threshold & Baseline Framework

Define KPI, baseline and threshold semantics without fabricating catalogues or numbers.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-001/003.

6.0 Salient Things to Note

KPI threshold breach ≠ proven violation; thresholds require governed sources — Table 9 fixes the identity/versioning/provenance structure every rule and KPI must carry, while deliberately leaving every concrete threshold value to Human Review rather than inventing one.

6.1 Breach Is Not Proven Violation

KPI THRESHOLD BREACH ≠ PROVEN VIOLATION. A KPI, its baseline, its threshold, its comparison window and its confidence/quality qualifiers are all defined structurally here; no specific KPI catalogue, numeric threshold or baseline value is invented — every concrete number requires a governed source.

6.2 KPI Catalogue and Thresholds

The KPI catalogue and its thresholds are Human Review decisions per module, informed by the domain-specific chapters (11-20).

6.3 Rule/KPI Governance Matrix (directive item 56)

The matrix below maps rule/KPI identity to owner-authority reference, versioning, inputs, thresholds/baselines, exceptions and provenance requirements.

Table 9. Rule/KPI Governance Matrix.

Element	Identity/Reference	Owner-Authority Reference	Version/Effective Time	Inputs	Thresholds/Baselines	Exceptions	Provenance
Domain rule (Ch.5)	Rule ID	Accountable Authority Class (to be assigned)	Versioned, effective-dated	Module-specific canonical fields (Ch.4)	N/A (rule logic, not a numeric KPI)	Recorded exception with reason	Full evaluation provenance
KPI (this chapter)	KPI ID	Accountable Authority Class (to be assigned)	Versioned, effective-dated	Module-specific canonical fields (Ch.4)	Not invented here — Human Review per module	Recorded exception with reason	Full computation provenance
Baseline (this chapter)	Baseline ID linked to KPI	Accountable Authority Class (to be assigned)	Versioned, windowed	Historical KPI values	Comparison window not invented here — Human Review	Recorded exception with reason	Full computation provenance

6.R RACI / Control Ownership

Table 10. Chapter 6 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve KPI catalogue/thresholds	Accountable Authority Class (to be assigned)	KPI/Threshold Framework Owner (to be assigned)	Module-specific owners (Ch.11-20)	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-6-1: Confirm the KPI catalogue and thresholds.
- Traceability: NCIE-001/003.

Anomaly, Risk & Scoring Framework

Define anomaly and risk-score semantics.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-007/008/010.

7.0 Salient Things to Note

ANOMALY ≠ FRAUD; RISK SCORE ≠ GUILT/LIABILITY — this chapter's score object always carries an explanation and calibration reference so a number is never presented without the context needed to interpret it responsibly.

7.1 Anomaly Is Not Fraud, Score Is Not Guilt

ANOMALY ≠ FRAUD, and RISK SCORE ≠ GUILT / LIABILITY. Every score carries a feature reference, a model/logic version, a confidence value, an explanation reference and a calibration reference — a high score is a prioritization signal for human review, never a substantiated conclusion.

7.2 Prohibited Autonomous Consequences

A score alone never triggers an autonomous consequential action (AUTOMATION ≠ UNBOUNDED CONSEQUENTIAL ACTION, cross-referenced from front matter §0.4) — any consequential action remains subject to NCIE-008/009/011 controls. Approved score uses and prohibited automation are a Human Review decision.

7.R RACI / Control Ownership

Table 11. Chapter 7 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve score uses/prohibited automation	Accountable Authority Class (to be assigned)	Anomaly/Risk Scoring Owner (to be assigned)	NCIE-007 model owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-7-1: Confirm approved score uses and prohibited automation.

Traceability: NCIE-007/008/010.

Alert, Situation & Escalation Framework

Define the Alert/Situation lifecycle.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-001/002/005/012.

8.0 Salient Things to Note

ALERT ≠ FINDING; urgency ≠ evidentiary certainty — this chapter tracks these as two independent fields on every Alert/Situation so a high-urgency item is never read as a high-confidence conclusion, or vice versa.

8.1 Alert Is Not Finding

ALERT ≠ FINDING. Every trigger produces an Alert or, where composite, a Situation, each with a severity, a confidence value, and — critically — urgency and evidentiary certainty tracked as two separate attributes so neither is mistaken for the other.

8.2 Acknowledgement, Escalation, Suppression and Closure

Acknowledgement, escalation, suppression and closure are each governed, attributable actions. Severity and escalation taxonomy specifics are a Human Review decision.

8.R RACI / Control Ownership

Table 12. Chapter 8 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve severity/escalation taxonomy	Accountable Authority Class (to be assigned)	Alert/Situation Owner (to be assigned)	NCIE-012 notification-UX owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-8-1: Confirm severity/escalation taxonomy.

Traceability: NCIE-001/002/005/012.

Case / Investigation Handoff

Define handoff into governed case/investigation workflows.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001/010/012.

9.0 Salient Things to Note

Analytics may trigger review but cannot autonomously create institutional Findings/Decisions — this chapter's case candidate is always a referral awaiting governed acceptance, never a self-opened case.

9.1 Analytics Cannot Autonomously Create Institutional Findings

Analytics may trigger review but cannot autonomously create institutional Findings or Decisions — a case candidate is a referral, not an opened case, until a governed NCIE-008 acceptance act creates it.

9.2 Referral Provenance

Every case candidate carries its Evidence gaps, referral basis, assignment and status provenance. Case-creation and referral authority is a Human Review decision.

9.R RACI / Control Ownership

Table 13. Chapter 9 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve case-creation/referral authority	Accountable Authority Class (to be assigned)	Case Handoff Owner (to be assigned)	NCIE-010 provenance owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-9-1: Confirm case-creation/referral authority.

Traceability: NCIE-001/010/012.

Cross-Domain Correlation & Fusion

Define governed cross-domain correlation between modules.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003/006/009/010.

10.0 Salient Things to Note

CORRELATION ≠ CAUSATION and cross-domain correlation ≠ disclosure authority — Table 14 fixes an explicit prohibited-disclosure-expansion rule for every correlation class so linking domains never becomes a backdoor around a source domain's own authorization boundary.

10.1 Correlation Is Not Causation or Disclosure Authority

CORRELATION ≠ CAUSATION, and CROSS-DOMAIN CORRELATION ≠ CROSS-DOMAIN DISCLOSURE AUTHORITY. Linking entities or time windows across modules (e.g., Network + Mobile Money) shows that a relationship was computed; it neither proves one caused another nor authorizes disclosing data from one domain to a consumer who lacks authorization for it.

10.2 Contradictions and Purpose

Cross-domain correlation preserves contradictions rather than resolving them silently, carries a stated purpose, and remains bound by NCIE-009 authorization at every source it touches. Permitted cross-domain purposes are a Human Review decision.

10.3 Cross-Domain Correlation Matrix (directive item 57)

The matrix below fixes permitted correlation inputs, purposes, authorization boundaries and prohibited disclosure expansion.

Table 14. Cross-Domain Correlation Matrix.

Correlation Class	Permitted Inputs	Purpose Basis	Authorization Boundary	Prohibited Disclosure Expansion
Entity correlation (e.g., subscriber across Network + Mobile Money)	Entity identifiers governed by NCIE-003	Stated analytical purpose (Human Review to confirm scope)	NCIE-009 current authorization at each source domain	Correlation never grants access to a source domain the requester was not already authorized to see
Time correlation (e.g., traffic anomaly + revenue variance in the same window)	Time-series across modules (Ch.20)	Stated analytical purpose	NCIE-009 current authorization at each source domain	Same as above
Composite/REWS correlation (Ch.17)	Leading indicators across modules	Early-warning purpose (Ch.17)	NCIE-009 current authorization at each source domain	Same as above

10.R RACI / Control Ownership

Table 15. Chapter 10 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve permitted	Accountable	Cross-Domain	NCIE-009	Assurance/Audit

cross-domain purposes	Authority Class (to be assigned)	Correlation Owner (to be assigned)	authorization owner	(NCIE-008)
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Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR13-10-1: Confirm permitted cross-domain correlation purposes.

Traceability: NCIE-003/006/009/010.

Network Performance & Quality Module

Expand Network Performance & Quality logic using only source-supported concepts.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-001/003/011.

11.0 Salient Things to Note

Performance degradation ≠ regulatory breach — this chapter's degradation and outage states are always presented as operational analytics, never as a pre-established regulatory violation.

11.1 Degradation Is Not Breach

Performance degradation is an analytical/operational state, not a regulatory breach — this module uses only source-supported coverage, availability, quality, outage and topology concepts, applying the Chapter 2 Domain-State Matrix and the Chapter 6 KPI/threshold framework without inventing a KPI catalogue or regulatory threshold here.

11.2 Trend and Geography

Trend and geographic presentation follow Chapter 19-20's geospatial/temporal rules; the specific KPI catalogue and any regulatory thresholds are a Human Review decision.

11.R RACI / Control Ownership

Table 16. Chapter 11 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve KPI catalogue/regulatory thresholds	Accountable Authority Class (to be assigned)	Network Performance Module Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [LEGAL/POLICY CONFIRMATION REQUIRED] (Blocking) HR13-11-1: Confirm KPI catalogue and regulatory thresholds.

Traceability: NCIE-001/003/011.

Traffic & Usage Intelligence Module

Expand Traffic & Usage logic for volumes, utilization, growth and anomalies.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001/003.

12.0 Salient Things to Note

Traffic anomaly ≠ misconduct; interpret with temporal/network/data-quality context — this chapter never surfaces a raw volume anomaly without the contextual fields needed to judge whether it is a normal seasonal pattern.

12.1 Anomaly Is Not Misconduct

A traffic anomaly (unusual volume, routing pattern, or seasonal deviation) is interpreted with temporal, network and data-quality context (Chapters 4, 20, 29) before any escalation — it is never presented as misconduct on its own.

12.2 Classification Specifics

Volume/utilization/growth/seasonality classifications and their thresholds are a Human Review decision.

12.R RACI / Control Ownership

Table 17. Chapter 12 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve classifications/thresholds	Accountable Authority Class (to be assigned)	Traffic & Usage Module Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-12-1: Confirm classifications and thresholds.

Traceability: NCIE-001/003.

Revenue & Financial Assurance Module

Expand Revenue & Financial Assurance reconciliation and variance logic.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001/003/010.

13.0 Salient Things to Note

Revenue variance ≠ fraud/liability; reconciliation gaps require Evidence/investigation — this chapter never labels a variance finding as fraud without a governed investigation producing that conclusion.

13.1 Variance Is Not Fraud or Liability

Revenue variance and leakage indicators are reconciliation-analysis outputs, not fraud or liability findings — a reconciliation gap requires Evidence and, where warranted, investigation (Chapter 9, 21) before any institutional conclusion attaches to it.

13.2 Source Precedence and Materiality

Which revenue source takes precedence when sources disagree, and what materiality threshold warrants escalation, are Human Review decisions — no materiality limit is invented here.

13.R RACI / Control Ownership

Table 18. Chapter 13 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve source precedence/materiality	Accountable Authority Class (to be assigned)	Revenue Assurance Module Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [LEGAL/POLICY CONFIRMATION REQUIRED] (Blocking) HR13-13-1: Confirm source precedence and materiality.

Traceability: NCIE-001/003/010.

Mobile Money Intelligence Module

Expand Mobile Money pattern, velocity, concentration and risk logic under strict privacy/purpose constraints.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001/003/009/010.

14.0 Salient Things to Note

Transaction anomaly/risk ≠ fraud; privacy/purpose remain strict — this chapter’s risk indicators are gated by the same NCIE-009 purpose-limitation check the underlying transaction data itself requires, with no relaxation for analytical convenience.

14.1 Anomaly Is Not Fraud, Privacy Is Strict

Transaction anomaly and risk indicators remain analytical states (ANOMALY ≠ FRAUD); mobile-money data is highly sensitive, so this module applies NCIE-009 purpose limitation and protected-identity controls at every stage — visibility of a pattern never itself justifies broader access.

14.2 Permitted Purposes

Permitted purposes for velocity/concentration/anomaly analysis and the risk indicators used are Human Review decisions, given the sensitivity of the underlying data.

14.R RACI / Control Ownership

Table 19. Chapter 14 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve permitted purposes/risk indicators	Accountable Authority Class (to be assigned)	Mobile Money Module Owner (to be assigned)	NCIE-009 privacy owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR13-14-1: Confirm permitted purposes and risk indicators.

Traceability: NCIE-001/003/009/010.

SIM / Subscriber Intelligence Module

Expand SIM/subscriber lifecycle, linkage and registration-quality logic.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-001/003/009.

15.0 Salient Things to Note

Identity linkage ≠ wrongdoing; protected identity remains governed — this chapter’s linkage output is treated as an analytical relationship record subject to the same protected-identity controls as any other SIM/subscriber field.

15.1 Linkage Is Not Wrongdoing

Identity linkage (associating SIMs, devices or subscribers) is an analytical relationship, not a finding of wrongdoing — protected identity remains governed by NCIE-009 throughout SIM lifecycle, linkage, registration-quality and pattern/exception analysis.

15.2 Identity-Use Purposes

Which purposes justify identity-linkage analysis, and which roles may access it, are Human Review decisions given the protected-identity sensitivity involved.

15.R RACI / Control Ownership

Table 20. Chapter 15 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve identity-use purposes/access	Accountable Authority Class (to be assigned)	SIM/Subscriber Module Owner (to be assigned)	NCIE-009 protected-identity owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR13-15-1: Confirm identity-use purposes and access.
- Traceability: NCIE-001/003/009.

Anti-Fraud & Abuse Intelligence Module

Expand fraud/abuse detection support logic.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001/007/008/010.

16.0 Salient Things to Note

ANOMALY ≠ FRAUD; indicators support review, not autonomous accusation/sanction — this chapter's referral object always routes to a human reviewer and never triggers a consequential action on its own.

16.1 Indicators Support Review, Not Accusation

ANOMALY ≠ FRAUD. Typology references, indicators, correlation, scores, alerts and referrals support human review — none of them constitutes an autonomous accusation or sanction. A referral surfaces Evidence gaps for investigation rather than asserting a conclusion.

16.2 Typology and Score Governance

The fraud typology reference and score-governance model are Human Review decisions; no typology catalogue is invented here.

16.R RACI / Control Ownership

Table 21. Chapter 16 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve typologies/score governance	Accountable Authority Class (to be assigned)	Anti-Fraud Module Owner (to be assigned)	NCIE-007 model owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [LEGAL/POLICY CONFIRMATION REQUIRED] (Blocking) HR13-16-1: Confirm typologies and score governance.

Traceability: NCIE-001/007/008/010.

REWS / Early-Warning Module

Expand regulatory early-warning and Situation-formation logic.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-001/005/010.

17.0 Salient Things to Note

Early warning is anticipatory intelligence, not a Finding; uncertainty/gaps remain visible — this chapter's composite signal always surfaces its underlying gaps rather than presenting a falsely complete picture.

17.1 Early Warning Is Anticipatory, Not a Finding

Early warning is anticipatory intelligence — leading indicators, trend breaks and composite signals feed into Situation formation (Chapter 8), but none of them constitutes a Finding. Uncertainty and gaps remain visible rather than smoothed away for a cleaner-looking signal.

17.2 Indicator Catalogue and Escalation

The leading-indicator catalogue and escalation rules are Human Review decisions.

17.R RACI / Control Ownership

Table 22. Chapter 17 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve indicator catalogue/escalation	Accountable Authority Class (to be assigned)	REWS Module Owner (to be assigned)	NCIE-005 behavioral owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-17-1: Confirm indicator catalogue and escalation.

Traceability: NCIE-001/005/010.

Operator / Entity Profile & Comparative Intelligence

Define governed operator/entity comparisons.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-003/012.

18.0 Salient Things to Note

Rank/outlier ≠ wrongdoing; comparisons normalize scope/time/quality — this chapter’s benchmark result is never presented without the normalization basis that makes the comparison meaningful.

18.1 Rank Is Not Wrongdoing

A rank or outlier position within a peer group is a comparative-analytics result, not a finding of wrongdoing — comparisons normalize for scope, time and data quality before any ranking is presented, and the normalization basis is always visible alongside the result.

18.2 Peer-Group and Benchmark Governance

Peer-group composition and benchmark governance are Human Review decisions.

18.R RACI / Control Ownership

Table 23. Chapter 18 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve peer-group/benchmark governance	Accountable Authority Class (to be assigned)	Comparative Intelligence Owner (to be assigned)	Reviewer	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-18-1: Confirm peer-group/benchmark governance.

Traceability: NCIE-003/012.

Geospatial & Topology Intelligence

Define geography and topology analysis across domain modules.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-003/009/012.

19.0 Salient Things to Note

Spatial/topology relation ≠ causality; sensitive layers remain authorized — this chapter never presents geographic proximity as proof of a causal or operational relationship between entities.

19.1 Spatial Relation Is Not Causality

Proximity, clustering and coverage layers show spatial relationships; a spatial relation is never presented as a causal one (CORRELATION ≠ CAUSATION applies spatially too). Temporal maps combine geography with time without collapsing either dimension.

19.2 Sensitive Layers

Some geospatial layers may be sensitive (e.g., precise location data); access to them follows NCIE-009 authorization, and which layers/analyses require heightened access is a Human Review decision.

19.R RACI / Control Ownership

Table 24. Chapter 19 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve sensitive-layer access/analyses	Accountable Authority Class (to be assigned)	Geospatial Intelligence Owner (to be assigned)	NCIE-009 classification owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR13-19-1: Confirm sensitive-layer access and analyses.

Traceability: NCIE-003/009/012.

Temporal, Trend & Forecast Interpretation

Define trend and forecast semantics.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-003/006/007.

20.0 Salient Things to Note

Forecast ≠ fact; late/revised data and historical baselines stay explicit — this chapter's forecast object always carries a confidence interval and is never rendered with the same visual certainty as an observed value.

20.1 Forecast Is Not Fact

A forecast is a model-derived projection, never presented as an established fact — observed time, effective time, seasonality, revision history and confidence intervals accompany every forecast so a reader can judge how much to rely on it.

20.2 Late/Revised Data and Decision-Time Comparison

Late and revised data remain explicit rather than silently folded into a smoothed series; historical baselines used for Decision-Time comparison follow the Chapter 16 (NCIE-010) reconstruction rules. Forecast use restrictions are a Human Review decision.

20.R RACI / Control Ownership

Table 25. Chapter 20 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve forecast use/restrictions	Accountable Authority Class (to be assigned)	Temporal/Forecast Owner (to be assigned)	NCIE-007 model owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-20-1: Confirm forecast use and restrictions.
- Traceability: NCIE-003/006/007.

Domain Evidence & Provenance Integration

Define module Evidence and provenance use.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-010.

21.0 Salient Things to Note

ANALYTICAL SIGNAL ≠ EVIDENCE; Evidence status comes only through NCIE-010 lifecycle — Table 26 fixes an explicit non-Evidence state for every analytical output class so none can be read as self-certifying Evidence.

21.1 Analytical Signal Remains Non-Evidence

ANALYTICAL SIGNAL ≠ EVIDENCE. Evidence status comes only through the NCIE-010 lifecycle — a domain module can link to registered Evidence, propose a candidate for registration, or flag a gap, but it cannot declare its own output to be Evidence.

21.2 Provenance, Contradictions and Decision-Time Packages

Domain Evidence links preserve provenance, contradictions, gaps, versions and custody exactly as NCIE-010 defines them, and assemble into a Decision-Time package (NCIE-010 Ch.16) where a Finding/Decision requires one. Default domain Evidence-package content is a Human Review decision.

21.3 Domain Evidence Matrix (directive item 58)

The matrix below maps analytical output classes to their Evidence relationship, explicitly including the non-Evidence state.

Table 26. Domain Evidence Matrix.

Analytical Output Class	Evidence Relationship	Provenance Reference	Gap/Contradiction Handling	Explicit Non-Evidence State
Raw observation (Ch.4)	May become a source for Evidence registration (NCIE-010 Ch.4)	Acquisition provenance	Gaps flagged, not silently filled	Non-Evidence until registered
Rule/KPI outcome (Ch.5-6)	References Evidence used as rule input where applicable	Evaluation provenance	Contradicting inputs preserved (NCIE-010 Ch.10-11)	The outcome itself is not Evidence
Anomaly/risk score (Ch.7)	May prompt Evidence-gap identification for investigation	Model/feature provenance	Score uncertainty preserved	The score itself is not Evidence
Alert/Situation (Ch.8)	References supporting analytical outputs, not Evidence directly	Trigger provenance	Contradicting alerts preserved, not merged	The alert itself is not Evidence
Case candidate (Ch.9)	Assembles a Decision-Time Evidence package where accepted	Referral provenance	Evidence gaps explicitly listed	The candidate itself is not a Finding

21.R RACI / Control Ownership

Table 27. Chapter 21 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve default domain Evidence-package content	Accountable Authority Class (to be assigned)	Domain Evidence Integration Owner (to be assigned)	NCIE-010 Evidence architecture owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-21-1: Confirm domain Evidence packages.

Traceability: NCIE-010.

ARGUS Domain Assistance Boundary

Define ARGUS assistance within domain modules.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-005/012.

22.0 Salient Things to Note

ARGUS assistance ≠ domain authority — this chapter's ARGUS interactions are bounded to summary, comparison and escalation, never a domain decision made on ARGUS's own output.

22.1 Assistance Is Not Authority

ARGUS ASSISTANCE ≠ DOMAIN AUTHORITY. ARGUS may summarize, compare, surface Evidence gaps, answer questions, propose actions and escalate within domain modules — none of these interactions grants ARGUS domain decision authority, which remains with the governed institutional processes this document defers to.

22.2 Provenance and Limits

Every ARGUS domain interaction preserves provenance to the underlying analytical outputs it references. Domain-assistance limits (which questions ARGUS may answer autonomously versus escalate) are a Human Review decision, bounded by NCIE-005's own authority limits.

22.R RACI / Control Ownership

Table 28. Chapter 22 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve domain assistance limits	Accountable Authority Class (to be assigned)	ARGUS Domain Assistance Owner (to be assigned)	NCIE-005 behavioral owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-22-1: Confirm domain assistance limits.
- Traceability: NCIE-005/012.

Agent & AI Use Within Domain Modules

Define bounded Agent and model use within domain modules.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-007/008/009.

23.0 Salient Things to Note

MODEL/AGENT OUTPUT ≠ FINDING/DECISION; VPF validation ≠ domain approval — this chapter never grants a domain-specific exception to the NCIE-007/008 boundaries that already govern every Agent/model output elsewhere in the platform.

23.1 Machine Output Never Becomes Finding or Decision

MODEL/AGENT OUTPUT ≠ FINDING/DECISION, and VPF validation ≠ domain approval. Agent tasks, model routes, Context references, result packages and VPF dispositions used within domain modules follow the same NCIE-007/008 boundaries as any other Agent/model use — a domain-specific application does not relax them.

23.2 Approved Domain AI/Agent Uses

Which specific Agent/model tasks are approved for use within each domain module, and what human review each requires, is a Human Review decision.

23.R RACI / Control Ownership

Table 29. Chapter 23 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve domain AI/Agent uses	Accountable Authority Class (to be assigned)	Domain AI/Agent Use Owner (to be assigned)	NCIE-007 orchestration owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-23-1: Confirm approved domain AI/Agent uses.

Traceability: NCIE-007/008/009.

Domain Workflow & Human Review Integration

Define module review and approval workflows.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-008/012.

24.0 Salient Things to Note

WORKFLOW STATE ≠ HUMAN APPROVAL — this chapter's review queue never treats a UI-level status change (e.g., moving an item to 'reviewed') as the governed approval act itself.

24.1 Workflow State Is Not Approval

WORKFLOW STATE ≠ HUMAN APPROVAL. A review queue, an assignment, or a UI status change is not itself an approval — approval requires the governed NCIE-008 decision-right act, evidenced and attributable to an accountable holder.

24.2 Review Queue and Dissent

Review queues, assignment, Evidence packages, approve/reject/dissent recording and closure follow NCIE-008 Ch.9-10 discipline. Review authorities and their module-specific variants are a Human Review decision.

24.R RACI / Control Ownership

Table 30. Chapter 24 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve review authorities/variants	Accountable Authority Class (to be assigned)	Domain Workflow Owner (to be assigned)	NCIE-008 Ch.9-10 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-24-1: Confirm review authorities and variants.

Traceability: NCIE-008/012.

Domain API & Connector Integration

Define consumption of NCIE-011 API and connector contracts.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-011.

25.0 Salient Things to Note

Connector availability ≠ source truth; no parallel integration semantics — this chapter consumes NCIE-011's own async/error/reconciliation states verbatim rather than defining a second, competing set.

25.1 Availability Is Not Truth

CONNECTOR AVAILABILITY ≠ SOURCE TRUSTWORTHINESS. A domain module consumes NCIE-011 APIs, connectors, events, schemas, reconciliation and async-state handling exactly as defined there — it does not define a parallel integration semantic layer of its own.

25.2 Standard versus Bespoke Connectors

Which domain connectors are standard, reusable NCIE-011 patterns versus module-specific bespoke integrations is a Human Review decision (mirrors NCIE-011 Ch.16 HR11-16-1).

25.R RACI / Control Ownership

Table 31. Chapter 25 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve standard vs bespoke connectors	Accountable Authority Class (to be assigned)	Domain API/Connector Owner (to be assigned)	NCIE-011 connector owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Non-Blocking) HR13-25-1: Confirm standard versus bespoke connectors; does not block consumption of already-defined NCIE-011 contracts in the interim.

Traceability: NCIE-011.

Domain UX & Workspace Integration

Map module logic to NCIE-012 UX/workspace patterns.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-012.

26.0 Salient Things to Note

Domain UX renders semantics without changing rules/authority — this chapter never introduces a domain-specific visual treatment for a state (e.g., an Alert) that conflicts with NCIE-012's own state-matrix definition of it.

26.1 Rendering Without Redefining

Domain UX (dashboards, tables, charts/maps, alerts, cases, Evidence, ARGUS, filters) renders domain semantics using the NCIE-012 component library and state matrices — it does not change what a rule, KPI, alert or Evidence state means or which authorization governs it.

26.2 Module Workspace Layouts

Specific module workspace layouts (which panels a Network-module analyst sees by default versus a Mobile-Money analyst) are a Human Review decision, informed by NCIE-012 Chapter 3's persona/workspace model.

26.R RACI / Control Ownership

Table 32. Chapter 26 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve module workspace layouts	Accountable Authority Class (to be assigned)	Domain UX Integration Owner (to be assigned)	NCIE-012 workspace owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Non-Blocking) HR13-26-1: Confirm module workspace layouts; does not block the state-rendering rules already consumed from NCIE-012 in the interim.

Traceability: NCIE-012.

Security, Privacy & Purpose Limitation

Apply NCIE-009 controls to all domain processing.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-009/008.

27.0 Salient Things to Note

Analytical value does not override purpose, authorization or privacy — this chapter never carves out a domain-specific exception to an NCIE-009 control merely because the analysis would be more useful without it.

27.1 Analytical Value Does Not Override Purpose

Analytical value does not override purpose, authorization or privacy — every domain module's classification, purpose, protected-identity, masking, DLP and audit handling is the same NCIE-009 enforcement applied to every other surface in the platform, with no domain-specific relaxation.

27.2 High-Risk Access Classes

Which domain-analytical access patterns count as high-risk and require heightened controls (e.g., cross-domain correlation touching Mobile Money and SIM data together) is a Human Review decision.

27.R RACI / Control Ownership

Table 33. Chapter 27 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve high-risk access classes	Accountable Authority Class (to be assigned)	Domain Security/Privacy Owner (to be assigned)	NCIE-009 classification owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR13-27-1: Confirm high-risk access classes.

Traceability: NCIE-009/008.

Domain Audit, Explainability & Defensibility

Define audit and explainability requirements for rules and outputs.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-010/007/008.

28.0 Salient Things to Note

Explainability supports review but does not create Evidence/Finding — this chapter's rationale and provenance records are retained for defensibility purposes but are never cited as Evidence in their own right.

28.1 Explainability Supports Review, Not Evidence

Explainability supports review but does not create Evidence or a Finding — a rule version, its inputs, its outputs, its rationale, and any model/Agent provenance behind it are all retained and reproducible, feeding the NCIE-010 Ch.27 defensibility package without becoming Evidence on their own.

28.2 Minimum Explainability

Human actions on domain outputs (acknowledgement, override, escalation) are themselves audited. The minimum explainability required per module/output class is a Human Review decision.

28.R RACI / Control Ownership

Table 34. Chapter 28 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve minimum explainability requirements	Accountable Authority Class (to be assigned)	Domain Audit/Explainability Owner (to be assigned)	NCIE-010 defensibility owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-28-1: Confirm minimum explainability.
- Traceability: NCIE-010/007/008.

Data Quality, Missingness & Reconciliation

Define incomplete, late, conflicting and revised data behavior.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.
Upstream: NCIE-003/010/011.

29.0 Salient Things to Note

MISSING DATA ≠ ZERO VALUE; LATE DATA ≠ CURRENT TRUTH — every computation in this document is required to propagate an explicit missing/late qualifier rather than substituting a default value that could be mistaken for a real observation.

29.1 Missing Is Not Zero, Late Is Not Current

MISSING DATA ≠ ZERO VALUE, and LATE DATA ≠ CURRENT TRUTH. A missing field is rendered and computed as explicitly missing, never silently defaulted to zero or a placeholder; late-arriving data does not retroactively claim to represent the current state until it is actually reconciled.

29.2 Conflict, Revision and Qualification

Conflicting or revised data triggers a reconciliation process and a qualification flag on affected outputs. Quality thresholds and suppression rules (when an output is too low-quality to surface) are a Human Review decision.

29.R RACI / Control Ownership

Table 35. Chapter 29 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve quality thresholds/suppression	Accountable Authority Class (to be assigned)	Data Quality Owner (to be assigned)	NCIE-010/011 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR13-29-1: Confirm quality thresholds and suppression rules.

Traceability: NCIE-003/010/011.

Domain Configuration & Change Governance

Define evolution of rules, thresholds, KPIs and configuration.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-008/004/010.

30.0 Salient Things to Note

Configuration changes alter analytical outcomes and require versioning/impact/governance — this chapter never allows a threshold or rule change to take effect without a recorded version, impact assessment and approval.

30.1 Configuration Changes Alter Outcomes

A configuration change (a new threshold, a revised rule, an updated KPI window) alters analytical outcomes — every configuration item carries an identity, version, proposal, test result, approval, effective date, rollback path and provenance, following the same versioning discipline as NCIE-011 Ch.20 schema governance.

30.2 Impact Assessment

A configuration change requires an impact assessment before approval; config approval authority is a Human Review decision. No configuration authority is invented here absent an approved source.

30.R RACI / Control Ownership

Table 36. Chapter 30 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve configuration approval authority	Accountable Authority Class (to be assigned)	Domain Configuration Owner (to be assigned)	NCIE-004 engineering owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-30-1: Confirm configuration approval authority.

Traceability: NCIE-008/004/010.

Domain Performance, Scale & Degraded Operation

Define functional behavior under degraded dependencies.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-004/011/012.

31.0 Salient Things to Note

Degraded mode must not relax Evidence/authorization/decision rules; stale/partial state is explicit — this chapter's degraded-mode indicator is required output whenever staleness or partial data could otherwise mislead a reviewer.

31.1 Degradation Does Not Relax Governed Rules

Degraded mode (a source outage, a growing backlog, stale freshness) must not relax Evidence, authorization or decision rules — a degraded state is surfaced explicitly (NCIE-012 Ch.24) rather than silently producing lower-quality results that look identical to normal output.

31.2 Recovery and Reconciliation

Recovery from a degraded state triggers reconciliation (NCIE-010 Ch.26, NCIE-011 Ch.29) before affected outputs are treated as fully current again. Freshness and availability expectations are a Human Review decision — no numeric target is invented here.

31.R RACI / Control Ownership

Table 37. Chapter 31 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve freshness/availability expectations	Accountable Authority Class (to be assigned)	Domain Performance/Degradation Owner (to be assigned)	NCIE-004 engineering owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR13-31-1: Confirm freshness and availability expectations.

Traceability: NCIE-004/011/012.

Functional Module Testing & Scenario Validation

Define domain test classes without duplicating NCIE-016.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-016 downstream; NCIE-008 assurance.

32.0 Salient Things to Note

TEST SPECIFIED ≠ TEST EXECUTED; integration success ≠ domain interpretation validity — Table 38 specifies eight test classes distinct from NCIE-011's integration tests, and every status field is marked to prevent specification from being mistaken for validated domain logic.

32.1 Test Classes

Eight test classes are specified: rule, KPI, anomaly, alert, cross-domain, Evidence traceability, security, and degraded-data scenario tests.

Table 38. Domain test-class specification.

Test ID	Test Class	Verifies	Status
DOM-T1	Rule Test	Ch.5: rule evaluation produces correct, versioned, provenance-complete outcomes for known inputs	TEST SPECIFIED
DOM-T2	KPI Test	Ch.6: KPI/baseline computation matches expected values for known input sets	TEST SPECIFIED
DOM-T3	Anomaly Test	Ch.7: anomaly/risk scoring produces explainable, calibrated output for known scenarios	TEST SPECIFIED
DOM-T4	Alert Test	Ch.8: Alert/Situation triggers, severity and escalation behave correctly for known scenarios	TEST SPECIFIED
DOM-T5	Cross-Domain Test	Ch.10: correlation stays within authorized sources and preserves contradictions	TEST SPECIFIED
DOM-T6	Evidence Traceability Test	Ch.21: domain outputs correctly link to or withhold Evidence status per Table 26	TEST SPECIFIED
DOM-T7	Security Test	Ch.27: domain processing preserves NCIE-009 purpose/authorization/classification controls	TEST SPECIFIED
DOM-T8	Degraded-Data Scenario Test	Ch.29, 31: missing/late/degraded data produces explicit qualifiers, never a silent default	TEST SPECIFIED

32.2 Test-Status Integrity

Specifying a test is not executing it (TEST SPECIFIED ≠ TEST EXECUTED); integration success does not itself validate domain interpretation — a module that correctly consumes an API (NCIE-011's own test) may still compute an incorrect analytical result if its domain logic is wrong. For every test above: IMPLEMENTATION EVIDENCE AVAILABLE

= NO; TEST EXECUTED = NO; TEST PASSED = N/A; SCENARIO VALIDATED = PENDING; HUMAN/PRODUCTION ACCEPTANCE = PENDING. Detailed test-case authoring and execution are downstream in NCIE-016.

32.R RACI / Control Ownership

Table 39. Chapter 32 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Domain)	Consulted	Informed
Approve scenarios/test acceptance authority	Accountable Authority Class (to be assigned)	Domain Test Owner (to be assigned)	Assurance/Audit (NCIE-008, independent per Ch.6 SoD)	NCIE-016 downstream owner

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-32-1: Confirm scenarios and acceptance authority.

Traceability: NCIE-016 downstream; NCIE-008 assurance.

Human Review Register & Open Domain Decisions

Consolidate every unresolved domain-business decision exposed across Chapters 1-32 and Chapter 34 into one register: deduplicated, consistently classified, each stating precisely what it blocks. No item here manufactures closure of a decision the approved upstream chain, the approved NCIE-008–012 governance, security, integration and UX layers, or the approved NCIE-013 Skeletal v0.1 does not itself settle.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-008–012 Human Review registers, plus domain-business-specific unresolved items from Chapters 1-32.

33.0 Salient Things to Note

Deduplicate upstream questions; create HR13 items only for distinct domain-business decisions; precise blocking scope.

33.1 Purpose

This register exists so that no unresolved domain-business decision is silently settled by expansion prose, and so that no chapter's own Human Review Focus item is left outside the register it is meant to feed — a defect found and corrected in the companion NCIE-008 v1.0 document's initial build. Every item from Chapters 1-32 below was extracted programmatically from that chapter's own Human Review Focus section (§33.4). Chapter 33's own item (HR13-33-1) and Chapter 34's own item (HR13-34-1) are each defined once as a fixed value and rendered identically both here and in their own chapter's Human Review Focus bullet, so neither can be omitted from the register or drift from the bullet that also states it.

33.2 Summary

Table 40. Human Review Register summary (mechanically computed).

Classification	Count
ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED	3
INSTITUTIONAL CONFIRMATION REQUIRED	22
LEGAL/POLICY CONFIRMATION REQUIRED	3
RESOLVED BY UPSTREAM BASELINE	1
SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED	5
Total	34
— Blocking	30
— Non-Blocking	4

33.3 Complete Register

Table 41. Complete Human Review Register.

Revie w ID	Source Chapte r	Question	Classification	Blocking Scope	Upstream Cross- Reference	Status
HR13-1-1	Chapte r 1	Confirm the NCIE-013 acceptance/sign-off authority and module ownership. The formal title has	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm the NCIE-013 acceptance/sign-off authority and module ownership. The formal title has	N/A — distinct domain-business decision	Open

		been reconciled against the NCIE Documentation Suite Master Controlled Register v2.0 (FOR APPROVAL) §3 — entry 'Functional Modules Specification' — and the document-control title corrected accordingly (§0.1); this item remains open pending institutional confirmation of acceptance authority and module ownership, and pending separate approval of the Master Controlled Register v2.0 itself.		been reconciled against the NCIE Documentation Suite Master Controlled Register v2.0 (FOR APPROVAL) §3 — entry 'Functional Modules Specification' — and the document-control title corrected accordingly (§0.1); this item remains open pending institutional confirmation of acceptance authority and module ownership, and pending separate approval of the Master Controlled Register v2.0 itself.		
HR13-2-1	Chapter 2	Approve common domain-state vocabulary.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Approve common domain-state vocabulary.	N/A — distinct domain-business decision	Open
HR13-3-1	Chapter 3	Confirm common lifecycle exceptions; does not block the nine-stage baseline lifecycle already defined in the interim.	INSTITUTIONAL CONFIRMATION REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level domain operation	N/A — distinct domain-business decision	Open
HR13-4-1	Chapter 4	Confirm datasets and source precedence per module.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm datasets and source precedence per module.	N/A — distinct domain-business decision	Open
HR13-5-1	Chapter 5	Confirm rule-authoring/approval authority.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm rule-authoring/approval authority.	N/A — distinct domain-business	Open

					decision	
HR13-6-1	Chapter 6	Confirm the KPI catalogue and thresholds.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm the KPI catalogue and thresholds.	N/A — distinct domain-business decision	Open
HR13-7-1	Chapter 7	Confirm approved score uses and prohibited automation.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm approved score uses and prohibited automation.	N/A — distinct domain-business decision	Open
HR13-8-1	Chapter 8	Confirm severity/escalation taxonomy.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm severity/escalation taxonomy.	N/A — distinct domain-business decision	Open
HR13-9-1	Chapter 9	Confirm case-creation/referral authority.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm case-creation/referral authority.	N/A — distinct domain-business decision	Open
HR13-10-1	Chapter 10	Confirm permitted cross-domain correlation purposes.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm permitted cross-domain correlation purposes.	N/A — distinct domain-business decision	Open
HR13-11-1	Chapter 11	Confirm KPI catalogue and regulatory thresholds.	LEGAL/POLICY CONFIRMATION REQUIRED (Blocking)	Confirm KPI catalogue and regulatory thresholds.	N/A — distinct domain-business decision	Open
HR13-12-1	Chapter 12	Confirm classifications and thresholds.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm classifications and thresholds.	N/A — distinct domain-business decision	Open
HR13-13-1	Chapter 13	Confirm source precedence and materiality.	LEGAL/POLICY CONFIRMATION REQUIRED (Blocking)	Confirm source precedence and materiality.	N/A — distinct domain-business decision	Open
HR13-14-1	Chapter 14	Confirm permitted purposes and risk indicators.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm permitted purposes and risk indicators.	N/A — distinct domain-business decision	Open
HR13-15-1	Chapter 15	Confirm identity-use purposes and access.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm identity-use purposes and access.	N/A — distinct domain-business decision	Open
HR13-	Chapter	Confirm	LEGAL/POLICY	Confirm	N/A —	Open

16-1	r 16	typologies and score governance.	CONFIRMATION REQUIRED (Blocking)	typologies and score governance.	distinct domain-business decision	
HR13-17-1	Chapter 17	Confirm indicator catalogue and escalation.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm indicator catalogue and escalation.	N/A — distinct domain-business decision	Open
HR13-18-1	Chapter 18	Confirm peer-group/benchmark governance.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm peer-group/benchmark governance.	N/A — distinct domain-business decision	Open
HR13-19-1	Chapter 19	Confirm sensitive-layer access and analyses.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm sensitive-layer access and analyses.	N/A — distinct domain-business decision	Open
HR13-20-1	Chapter 20	Confirm forecast use and restrictions.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm forecast use and restrictions.	N/A — distinct domain-business decision	Open
HR13-21-1	Chapter 21	Confirm domain Evidence packages.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm domain Evidence packages.	N/A — distinct domain-business decision	Open
HR13-22-1	Chapter 22	Confirm domain assistance limits.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm domain assistance limits.	N/A — distinct domain-business decision	Open
HR13-23-1	Chapter 23	Confirm approved domain AI/Agent uses.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm approved domain AI/Agent uses.	N/A — distinct domain-business decision	Open
HR13-24-1	Chapter 24	Confirm review authorities and variants.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm review authorities and variants.	N/A — distinct domain-business decision	Open
HR13-25-1	Chapter 25	Confirm standard versus bespoke connectors; does not block consumption of already-defined NCIE-011 contracts in the	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level domain operation	NCIE-011 HR11-16-1 (standard vs bespoke domain connectors) — same institution	Open

		interim.				al question, instantiated identically in NCIE-011 Ch.16; resolving HR11-16-1 resolves this item too.	
HR13-26-1	Chapter 26	Confirm module workspace layouts; does not block the state-rendering rules already consumed from NCIE-012 in the interim.	INSTITUTIONAL CONFIRMATION REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level domain operation	N/A — distinct domain-business decision	Open	
HR13-27-1	Chapter 27	Confirm high-risk access classes.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm high-risk access classes.	N/A — distinct domain-business decision	Open	
HR13-28-1	Chapter 28	Confirm minimum explainability.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm minimum explainability.	N/A — distinct domain-business decision	Open	
HR13-29-1	Chapter 29	Confirm quality thresholds and suppression rules.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm quality thresholds and suppression rules.	N/A — distinct domain-business decision	Open	
HR13-30-1	Chapter 30	Confirm configuration approval authority.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm configuration approval authority.	N/A — distinct domain-business decision	Open	
HR13-31-1	Chapter 31	Confirm freshness and availability expectations.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm freshness and availability expectations.	N/A — distinct domain-business decision	Open	
HR13-32-1	Chapter 32	Confirm scenarios and acceptance authority.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm scenarios and acceptance authority.	N/A — distinct domain-business decision	Open	
HR13-33-1	Chapter 33	Review this consolidated	INSTITUTIONAL CONFIRMATION REQUIRED	Review this consolidated	N/A — distinct	Open	

		register and decide, for each Blocking item, whether it must be resolved before NCIE-013 v1.0 may be approved or may instead remain open under an interim design default.	(Blocking)		register and decide, for each Blocking item, whether it must be resolved before NCIE-013 v1.0 may be approved or may instead remain open under an interim design default.	domain-business decision	
HR13-34-1	Chapter 34	Approve the NCIE-013 skeletal structure and authorize controlled full expansion — already satisfied by the approved NCIE-013 Skeletal v0.1 and the production directive that authorized this v1.0 expansion; preserved here as historical record, not a live open item.	RESOLVED BY UPSTREAM BASELINE (Non-Blocking)		Non-blocking — does not gate chapter-level domain operation	N/A — distinct domain-business decision	Resolved by upstream baseline

33.4 Mechanical Verification

32 of these 34 items were recomputed programmatically from the Human Review Focus bullet text of Chapters 1-32 rather than assembled by hand. The remaining 2 items are HR13-33-1 (this chapter's own item) and HR13-34-1 (Chapter 34's own item), each defined once and rendered identically in both this table and their own chapter's Human Review Focus bullet. 0 duplicate Review IDs; 0 orphan Human Review items — every Human Review ID appearing anywhere in this specification is reconciled into this register.

33.5 Upstream Deduplication (Directive Item 60)

One item, HR13-25-1, restates the identical institutional question NCIE-011 has already opened as HR11-16-1 (standard versus bespoke domain connectors); Table 41 cross-references it rather than treating it as an independent open question, per directive item 60. No other NCIE-013 item was found to duplicate an existing NCIE-008-012 governance, security, integration or UX question — each remaining item concerns a domain-business-specific decision that those documents do not themselves resolve.

33.6 Blocking Scope Discipline

Each Blocking item states exactly what it blocks — a specific rule catalogue, threshold, typology, ownership assignment or acceptance authority — rather than implying the whole of NCIE-013 or the whole approved upstream chain is blocked. Non-Blocking items are explicitly proposed defaults or confirmations that do not gate the corresponding chapter's domain operation in the interim.

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR13-33-1: Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-013 v1.0 may be approved or may instead remain open under an interim design default.

Traceability: All chapters above (mechanically reconciled register); NCIE-008–012 Human Review registers.

Traceability, Acceptance & Downstream Handover

Close traceability to the approved NCIE-001-012 foundation and the approved NCIE-013 Skeletal v0.1; define acceptance criteria and the downstream handover to NCIE-015/016.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: Approved NCIE-001-012 foundation.

34.0 Salient Things to Note

Skeleton approval authorizes expansion only; approval does not prove rule correctness, data availability, readiness or testing — §34.4's acceptance gate states this explicitly rather than leaving it implied.

34.1 Traceability Matrix

Every chapter of this document traces to an approved upstream authority or to the approved NCIE-013 Skeletal v0.1 itself. No chapter states a domain-business requirement without an identifiable source.

Table 42. Full traceability matrix.

NCIE-013 Chapter	Primary Upstream Authority	Relationship
Chapter 1	NCIE-001-012; NCIE-005/011/012 handovers	Mandate/domain-module boundary
Chapter 2	NCIE-001/003/008/010	Cross-domain functional doctrine
Chapter 3	NCIE-002/011/012	Module architecture & common lifecycle
Chapter 4	NCIE-003/010/011	Domain data inputs & canonical mapping
Chapter 5	NCIE-003/008	Domain rule & policy evaluation framework
Chapter 6	NCIE-001/003	KPI, threshold & baseline framework
Chapter 7	NCIE-007/008/010	Anomaly, risk & scoring framework
Chapter 8	NCIE-001/002/005/012	Alert, Situation & escalation framework
Chapter 9	NCIE-001/010/012	Case/investigation handoff
Chapter 10	NCIE-003/006/009/010	Cross-domain correlation & fusion
Chapter 11	NCIE-001/003/011	Network Performance & Quality module
Chapter 12	NCIE-001/003	Traffic & Usage Intelligence module
Chapter 13	NCIE-001/003/010	Revenue & Financial Assurance module
Chapter 14	NCIE-001/003/009/010	Mobile Money Intelligence module
Chapter 15	NCIE-001/003/009	SIM/Subscriber Intelligence module
Chapter 16	NCIE-001/007/008/010	Anti-Fraud & Abuse Intelligence module
Chapter 17	NCIE-001/005/010	REWS/Early-Warning module
Chapter 18	NCIE-003/012	Operator/Entity Profile & Comparative Intelligence
Chapter 19	NCIE-003/009/012	Geospatial & Topology Intelligence

Chapter 20	NCIE-003/006/007	Temporal, Trend & Forecast Interpretation
Chapter 21	NCIE-010	Domain Evidence & provenance integration
Chapter 22	NCIE-005/012	ARGUS domain assistance boundary
Chapter 23	NCIE-007/008/009	Agent & AI use within domain modules
Chapter 24	NCIE-008/012	Domain workflow & Human Review integration
Chapter 25	NCIE-011	Domain API & connector integration
Chapter 26	NCIE-012	Domain UX & workspace integration
Chapter 27	NCIE-009/008	Security, privacy & purpose limitation
Chapter 28	NCIE-010/007/008	Domain audit, explainability & defensibility
Chapter 29	NCIE-003/010/011	Data quality, missingness & reconciliation
Chapter 30	NCIE-008/004/010	Domain configuration & change governance
Chapter 31	NCIE-004/011/012	Domain performance, scale & degraded operation
Chapter 32	NCIE-016 (downstream); NCIE-008	Functional module testing & scenario validation
Chapter 33	All chapters above; NCIE-008-012 Human Review registers	Consolidated Human Review Register
Chapter 34	Approved NCIE-001-012; NCIE-013 Skeletal v0.1	Traceability/acceptance/handover

34.2 Chapter-Count and Salient-Note Mechanical Verification

A programmatic scan of this document confirms exactly 34 sequential 'NCIE-013 — Chapter N Expansion' labels, N = 1..34, each present exactly once, in order, with no gap, duplicate or unintended renaming (directive item 70). A second scan confirms an explicit '<N>.0 Salient Things to Note' heading present for all 34 chapters (directive item 71), rerun independently against this final document rather than assumed from the build script.

34.3 Skeleton-to-Full Fidelity Verification

Every approved Chapter Purpose, Salient Things to Note, Principal Component, Upstream Dependency/Authority and Human Review Focus from the approved NCIE-013 Skeletal v0.1 is substantively represented in the corresponding full chapter above (directive item 69); none is omitted, and none is conceptually inverted.

34.4 Acceptance Criteria and the Unresolved-Item Gate

Full approval of NCIE-013 v1.0 requires: (a) 34/34 chapter integrity confirmed; (b) 34/34 explicit Salient Things to Note subsections confirmed; (c) skeleton-to-full fidelity confirmed; (d) every Blocking Human Review item in Chapter 33 either resolved by an institutional decision or explicitly and knowingly deferred by the approving authority (including the formal-title reconciliation, §0.1, HR13-1-1); (e) the §32.1 test suite specified with no test falsely marked executed, passed, or scenario-validated; and (f) three-artifact consistency (this document, the Change Register and the Verification Report) confirmed. Approval of the v0.1 skeleton authorized this expansion only — it did not itself approve any rule catalogue, threshold, typology or ownership assignment proposed above. Domain-module approval under this chapter does not prove rule correctness, data availability, analytical validity or test execution.

34.5 Downstream Handover Register

NCIE-013 does not preempt detailed downstream specification. The table below states what is handed over and what remains downstream.

Table 43. Downstream handover register.

Downstream Document	What NCIE-013 Hands Over
NCIE-015 (deployment/operations)	Domain module operational responsibilities (RACI/control-ownership tables throughout) for handoff into deployment/operational runbooks; no runbook is specified here.
NCIE-016 (detailed testing)	The 8 domain test classes specified in §32.1 (Table 38); detailed test cases, execution and scenario validation remain downstream in NCIE-016 — this document specifies the tests, not their execution.

34.6 Readiness and Final Disposition

NCIE-013 Functional Modules Specification v1.0 — Full Expansion of Approved Skeletal v0.1 — is READY FOR HUMAN REVIEW. It is NOT marked Approved. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed rule catalogue, threshold, typology and ownership assignment deferred in Chapters 1-32 remains a proposal pending the Chapter 33 Human Review Register; every §32.1 test remains SPECIFIED, not executed or scenario-validated, pending NCIE-016.

Human Review Focus

- [RESOLVED BY UPSTREAM BASELINE] (Non-Blocking) HR13-34-1: Approve the NCIE-013 skeletal structure and authorize controlled full expansion — already satisfied by the approved NCIE-013 Skeletal v0.1 and the production directive that authorized this v1.0 expansion; preserved here as historical record, not a live open item.

Traceability: NCIE-001-012 approved foundation; approved NCIE-013 Skeletal v0.1.